Eiusdem
EXPERIMENTVM CIRCA NERVVM OPTICVM.

EXtat in Act. Lipſ. A. 1683. pag. 68. experimentum, quo demonſtratur, ſpatiolum eſſe in media choroidea (ſive, ut alii volunt, retina) ad imaginem obiectorum percipiendam plane ineptum, eoque D. Mariotte uſus eſt contra D. Pecquet in examinando primario viſus organo. Diſceptatum autem fuit a viris clariſſimis, an perforationi choroideæ a nervo optico, an vero plexui vaſculorum ſanguiferorum in retina conſpicuo tribuendum ſit, quod experimento monſtratur. vid. *Hiſt. Reg. Sc. Acad. Paris. conſcriptam a D. Du Hamel.* Ego autem putavi, id diiudicandum eſſe ex figura, ſitu & magnitudine loci illius extra radiorum actionem poſiti; poſſunt enim hæc omnia exactiſſime determinari ſequentem in modum.

Fig. 7. In pavimento (cuius parti aliquot pedes longæ ad meliorem experimenti ſucceſſum pannus niger inſterni poteſt) ponatur moneta argentea O: ducaturque recta OA; Dein accipiatur pendulum, cuius altera extremitas oculo dextro admoveatur, altera ad pavimentum non omnino pertingat. Tandem clauſo oculo ſiniſtro directaque conſtanter dextri acie in imam penduli partem, ita paulatim corpus verſus ſiniſtram promoveatur, ut penduli extremitas continuo ſit in recta OA, quæ dum fiunt, exacte

acte obſervetur locus, ubi obiectum O viſu perdi incipit. Sit ille in C; poſteaque continuando motum, accurate quoque notetur punctum A, in quo obiectum O iterum percipitur: Biſecta linea AC in D, ductaque perpendiculari BH, fiat in BH, quod ante in linea OA, id eſt, diligenter obſerventur puncta B & H tanquam limites intra quos cum exiſtit penduli extremitas, obiectum O viſui ſe ſurripit: linea BH denuo biſecetur in F linea perpendiculari GE, & in hac pari conditione quærantur puncta E & G; idem fiat in aliis lineis LP, MN &c. Atque per omnia iſta puncta ducatur linea curva HEBGH, quæ erit huius indolis, ut quamdiu oculus correſpondet puncto intra curvam, obiectum O nullo modo appareat, ſi extra curvam cadit axis viſionis, idem mox percipiatur. Dicam nunc, quæ ipſe experiundo obſervavi reductis ſingulis quantitatibus ad longitudinem penduli, quam voco 100. Inveni ergo OC=23, AC=10, BD=3, DH=13, & EG=14: Ex quibus liquet, curvam accedere ad ellipſin. Cuius centrum eſt in F, & cuius axis maior eſt ad axem minorem ut 8 ad 7.

Reſtat nunc indagandum, quomodo ex obſervatis his determinari poſſint ſitus, figura & magnitudo ſpatii illius ad viſum inutilis. Id vero nullo negotio obtinetur; cognitis enim lineis AC, EG, LP, MN, HB &c. determinantur lineæ analogæ in oculo. Quamvis autem BH maior ſit quam EG, fient tamen ipſarum analogæ in oculo æquales, adeo ut figura elliptica in circularem mutetur: & erit diameter circelli illius æqualis ſeptimæ parti diametri totius bulbi. Centrum eius diſtabit ſeptem vigeſimis quintis partibus diametri bulbi a puncto pupillæ op-

posito, eritque simul supra medium oculi paululum elevatum. Hæc omnia exacte cum nervo optico in oculo humano conveniunt adeo, ut iam sententia D. Mariotte sit hac in parte extra omne dubium posita.

Quod si ergo insertio nervi optici sit causa phænomeni illius a D. Mariotte observati, planum est, non debuisse nervum hunc directe e regione pupillæ in bulbum inseri: ita enim omnia obiecta in medio perforata nobis apparerent, illudque, in quod oculos defixos nunc habere solemus, non videremus: dum vero nervus opticus lateraliter bulbo insertus fuit, efficitur, ut illa tantum obiecta perdamus, quæ sub certo angulo ad latus sunt posita. Neque unquam unum idemque obiectum amborum oculorum sensui se surripere potest, si nervi optici a latere sint oculo implantati; id vero semper fieret, si essent pupillæ e diametro oppositi.

Alterum Corollarium, quod deduco ex eodem experimento, tale est. Non potuit nervus opticus aliter quam ad perpendiculum oculo inseri. Nam cum locus insertionis sit ad visum inutilis, natura in id intenta esse debuit, ut fieret inter omnes possibiles minimus: hocque obtinetur, dum perpendiculariter bulbum nervus ingreditur.

Quod si nervi recta ab originibus suis, scil. cruribus medullæ oblongatæ ad oculos tetendissent, necessario contra alterutrum præmissorum lemmatum peccatum fuisset; cum enim principium nervi optici sit fere e regione oculi constitutum, uti hic in A, manifestum est, esse nervum AB (quem perpendicularem pono ad sphæram seu bulbum oculi) e regione pupillæ positum. Sed si nervus

Fig. 8.

late-

fig: 1.

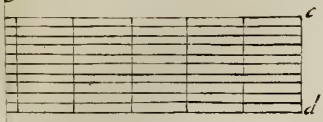
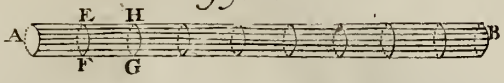


fig: 2.





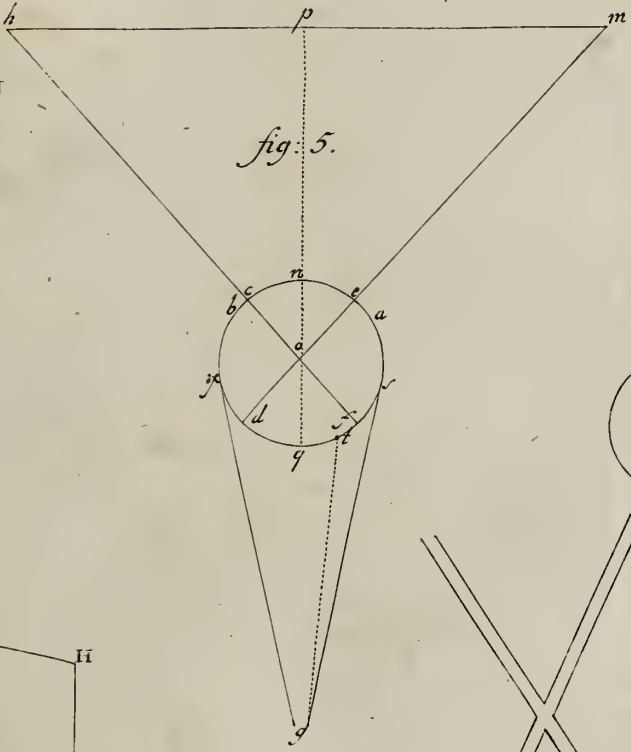
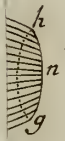


fig: 5.



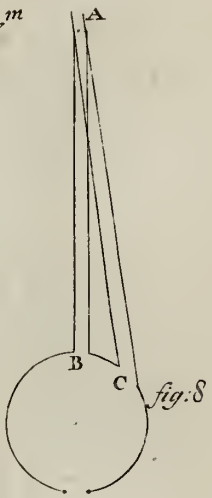
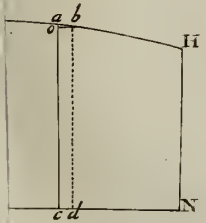


fig: 8



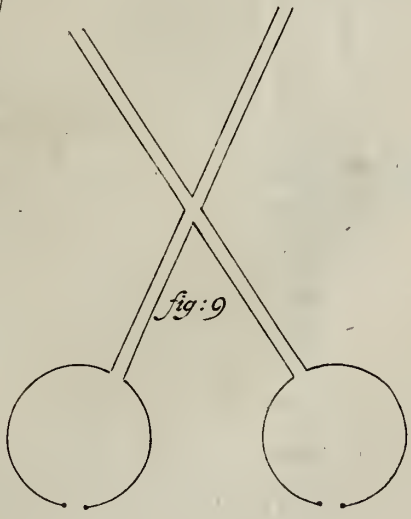


fig: 9

fig: 7



fig: 1



fig: 2.

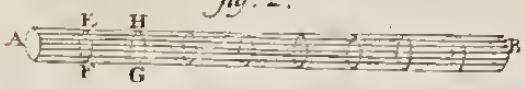
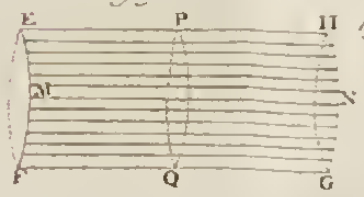


fig: 3.



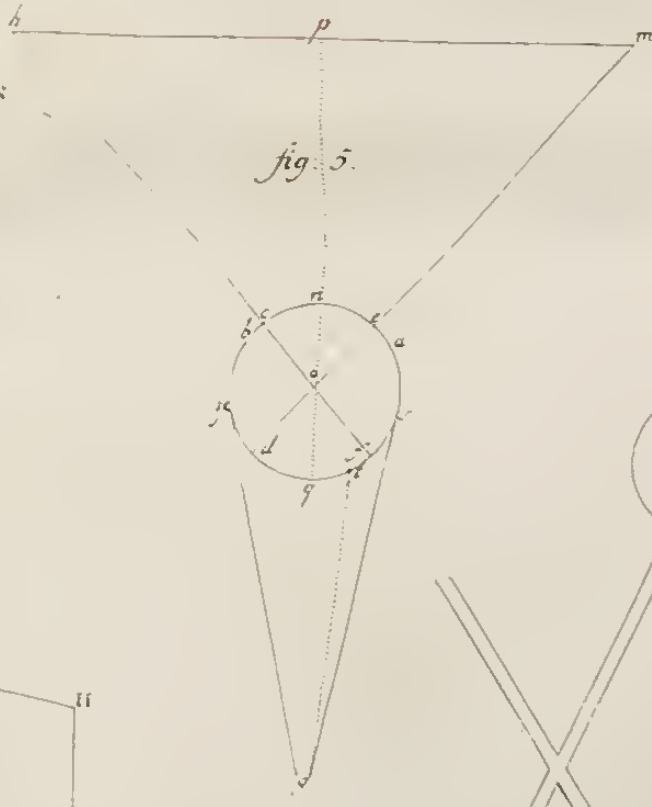


fig: 4



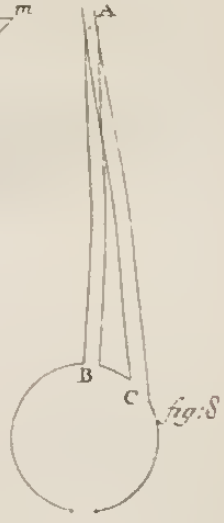
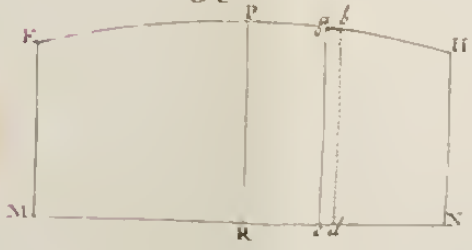
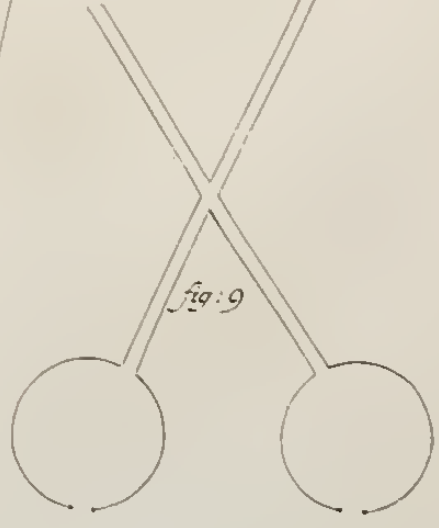


fig: 6.





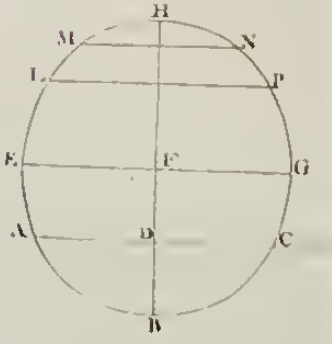


fig: 7



lateraliter in C fuiſſet oculo inſertus, eo ipſo obliquus evaſiſſet.

Dum vero nervi non recta tendunt ad oculum, ſed prius coniunguntur, utrique poſtulato ex aſſe ſatisfit, uti id ex ſola inſpectione figuræ nonæ apparet. *Fig.* 9.

Si quis exinde rationem coniunctionis nervorum opticorum deducere volet, ei facile ſuffragabor.

DE VARIIS BAROMETRIS SENSIBILIORIBVS, ET EORVM NOVA SPECIE, AC VSIBVS:

Auctore

G. B. Bülffingero.

Recensio Barometrorum.

I.

ORiginem Barometrorum & ſpecies expoſuit *Dalencéus* ſingulari libello. Pauca ex illo tempore nova acceſſerunt, ſed aliqua tamen, eaque digna Eruditorum attentione. Præcipua Virorum cura fuit, ut variationes altitudinum fiant *ſenſibiliores*; huic enim operæ incubuerunt *Carteſius*, *Hoockius*, *Hugenius*, *Amontonius*, *M. Aug.* 1726.